Jessica Reiff

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Summary

Electrical Engineering student with experience in embedded firmware, sensor fusion, robotics, and real-time control systems. Skilled in STM32 development, ROS2, embedded communications, and autonomous systems engineering.

Education

Iowa State University, Bachelor of Science in Electrical Engineering – Ames, IA **Expected Dec 2026**

- GPA: 3.87

Professional Experience

Embedded Intern, AthenaGTX – Johnston, IA **May 2025 – Aug 2025**

- Developed embedded C firmware for motion-tracking prototypes using BMI323 IMU sensors and BLE communication via BMD-350 modules.
- Updated legacy firmware by adding dual-color LED indicators and improving pump control behavior.
- Debugged hardware–firmware interactions using custom Android BLE test applications written in Java to validate communication and UI synchronization.
- Collaborated with hardware and mechanical teams on PCB development, system integration, and documentation.

MATLAB Student Ambassador, MathWorks – Ames, IA **Jan 2025 – Present**

- Coordinated workshops and seminars to demonstrate MATLAB's capabilities, enhancing students' technical skills.
- Provided peer support and troubleshooting for MATLAB-related issues and optimization best practices.
- Acted as liaison between MathWorks and university, increasing MATLAB adoption through targeted outreach.

Undergraduate Research Assistant, Iowa State College of Engineering – Ames, IA **Mar 2025 – Present**

- Develop autonomous driving algorithms for the F1TENTH platform using ROS2 in simulation and real-world autonomous racecar environments.
- Design and test path planning, obstacle avoidance, and control algorithms for autonomous navigation.
- Implement sensor fusion pipelines using LiDAR, IMU, and camera data for localization and perception in autonomous systems.

Academic Project

OSCAR – Open-Source Copter for Academic Research | Firmware & Controls Engineer **Jan 2026 – Dec 2026**

- Developed embedded firmware for STM32-based quadcopter flight controller hardware using SPI, I2C, UART, and USB communication protocols.
- Implemented quaternion-based orientation estimation and sensor fusion using IMU and magnetometer sensor data.
- Integrated ExpressLRS receiver communication for low-latency pilot control input.
- Configured and debugged real-time sensor pipelines and high-frequency flight control subsystems.
- Performed register-level peripheral configuration and hardware debugging using oscilloscopes and logic analyzers.

Leadership

Electrical Engineer Curriculum Chair, IEEE **Aug 2024 – Present**

- Attend ECPE Curriculum meetings as representative for Electrical Engineering students.
- Coordinate with faculty and peers to provide student feedback on curriculum updates.

Technical Skills

- **Programming Languages:** C, Embedded C, C++, Python, Java, MATLAB, VHDL, Verilog, C#
- **Embedded Robotics:** STM32, FreeRTOS, ROS2, Sensor Fusion, Quaternion Mathematics, Madgwick AHRS, Real-Time Systems, Embedded Firmware Development
- **Communication Protocols:** SPI, I2C, UART, BLE, USB
- **Tools & Software:** STM32CubeIDE, MATLAB, Simulink, Android Studio, KiCad, QuestaSim, Git
- **Hardware & Lab:** Oscilloscopes, Logic Analyzers, PCB Prototyping, Soldering, Circuit Debugging